

ELIA STOLK

027 · 265 · 0559 ◇ eliafolk9@gmail.com

Auckland, New Zealand

EDUCATION

University of Auckland

November 2027 (*Estimated completion date*)

Bachelor of Engineering (Honours), Electrical and Electronic Engineering

Cumulative GPA: 7.438

PROJECTS

For full list of projects visit <https://eliastolk.github.io/>

Capacitive Water Level Monitoring System (Team Lead)

May 2026

- Led a team of engineering students in the development of a capacitive water level monitoring system for agricultural water tanks.
- Designed and analysed sensing, infrared transmission, and analogue receiver subsystems using LTspice simulation and prototype testing.
- Evaluated design trade-offs involving signal integrity, component cost, bandwidth, noise immunity, and system reliability.
- Delivered a fully functional prototype and achieved the highest project report grade in the cohort (14.5/15).

Smart Energy Monitor

October 2025

- Designed dual-gain amplifier and signal-conditioning stages to measure AC voltage, current, and power on a custom SMT PCB
- Simulated circuit behaviour in LTSpice and Proteus to validate gain, filtering, and measurement accuracy prior to hardware testing
- Selected component values and configured an ATmega328P microcontroller to sample and compute voltage, current, power, and energy in real time
- Evaluated design trade-offs including measurement bandwidth, component cost, and noise immunity across the sensing front end

PROFESSIONAL EXPERIENCE

VAPO

Sales Representative

March 2024 - Present

Mount Wellington, Auckland

- Provide premium service in a specialist retail setting, optimising sales and customer satisfaction
- Apply product knowledge to identify customer needs and recommend suitable solutions
- Resolve device issues within company return policy guidelines

VAPO

Store Manager

February 2023 - February 2024

New Lynn, Auckland

- Managed inventory stock takes and store performance reporting
- Trained and coached team members through regular one-on-one sessions

TECHNICAL STRENGTHS

Design Software

Altium Designer, Quartus 2, Autodesk Inventor, LaTeX, R Studio

Simulation Software

Proteus 8 Professional, LTSpice, FEMM

Workshop Skills

3D Printing, Soldering, Health and Safety Protocols

Programming

MATLAB, C, Python, Java, Git, FPGA, Microchip Studio